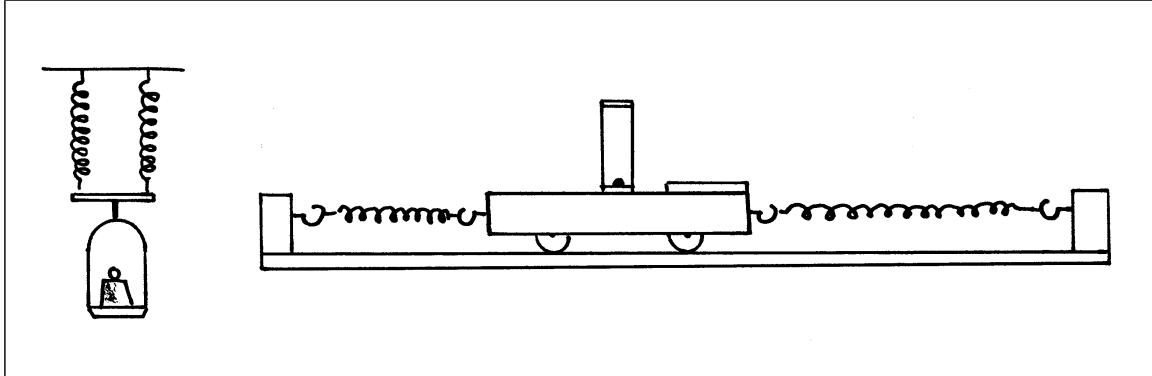


## 5: Measuring the model point mass

For an introduction we consider fig. 5.1 where it is shown how to determine the mass of a car without actually weighing it on a pair of scales.



**Figures 5.1a and b:** How to determine the mass  $m$  of the car without actually weighing it on a pair of scales.

Figure a shows a known mass<sup>1</sup>  $\mu$ , suspended in two identical springs. Determining the period  $T_\mu$  for an oscillation up and down, the force constant for the two springs together is given by

$$k = \frac{4\pi^2\mu}{T_\mu^2} \quad (\Leftrightarrow \mu = \frac{kT_\mu^2}{4\pi^2})$$

Figure b shows the car suspended between the same two springs as in fig. b. Determining the period  $T$  for an oscillation back and forth, the mass of the car can be found as

$$m = \frac{kT^2}{4\pi^2} = \mu \frac{T_\mu^2}{T^2}$$

With the method of fig. 5.1 in mind, it is intuitively fairly obvious that the model point mass on the merry-go-round can be determined as explained in the first of the two figures 5.2 below; i.e., fig. 5.2a. If  $m_e$  denotes the value found *without* a model, the value *with* a model of mass  $m_{\text{model}}$  and a counterbalance mass  $m_{\text{counter}}$  in the box opposite the protruding arm, should be

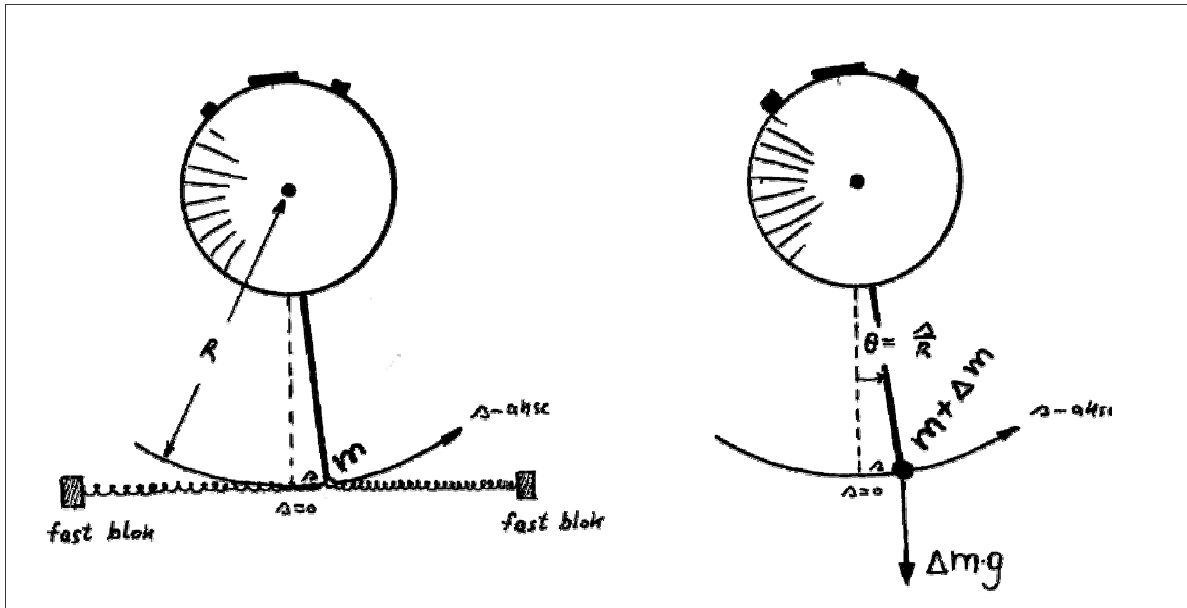
$$m = m_e + m_{\text{model}} + m_{\text{load}}; \quad m_{\text{load}} = \left(\frac{r_{\text{counter}}}{R}\right)^2 m_{\text{counter}} \quad (5.1)$$

(compare with (4.3)) where  $r_{\text{counter}}$  is the distance from the axis to  $m_{\text{counter}}$ . Here it agrees with our intuition that since  $r_{\text{counter}}$  is smaller than  $R$ ,  $m_{\text{counter}}$  should not be added with its full value in the same way as  $m_{\text{model}}$ . The reader who wants to understand why the reducing factor equals  $(r_{\text{counter}}/R)^2$  must look into the elementary theory of a rigid body's rotation about an axis. See appendix B.

<sup>1</sup> As is well known, the two springs themselves do actually contribute to the oscillating mass by  $\Delta\mu$ , equal to one third of their total mass. For very accurate measurements, the formulas for  $k$  and  $m$  should therefore be corrected to

$$k = \frac{4\pi^2(\mu + \Delta\mu)}{T_\mu^2} \quad (\Leftrightarrow \mu + \Delta\mu = \frac{kT_\mu^2}{4\pi^2}) \quad \text{and} \quad m + \Delta\mu = \frac{kT^2}{4\pi^2} = (\mu + \Delta\mu) \frac{T_\mu^2}{T^2}$$

If  $m$  and  $\mu$  are close to being equal, so are  $T$  and  $T_\mu$  the influence of  $\Delta\mu$  can be neglected.



**Figure 5.2 a and b:** In fig. a, the merry-go-round is mounted with the axis of rotation vertical, and one sees it from above. The model point is suspended between two springs so that it can oscillate along the  $s$ -axis with a small amplitude. Measuring the period  $T$  for this oscillation, one finds that the model point represents the inertial mass

$$m = \frac{kT^2}{4\pi^2} \quad (\text{or} \quad m = \mu \frac{T_\mu^2}{T^2})$$

where  $k$  is the force constant for the two springs together. ( The mass  $\mu$  and time  $T_\mu$  in the second expression are those appearing if the force constant has been determined as in fig. 5.1a ).

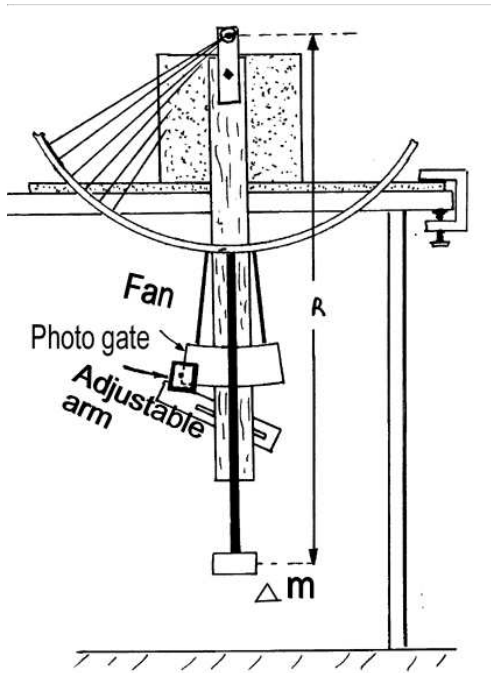
Fig. b illustrates an alternative method for determination of  $m$ . This is meant to be carried out just after the counterbalances have been adjusted ( see the subsection below ) to make the centre of mass coincide with the axis of rotation, and this axis is still fastened horizontally. Fixing an extra mass  $\Delta m$  in the model point as shown in fig. 5.3, we get a pendulum. Measuring period  $T$  when this swings back and forth with a small amplitude we then have

$$m = \Delta m \left( \frac{g}{4\pi^2 R} T^2 - 1 \right)$$

where  $g = 9.82 \frac{m}{s^2}$  is acceleration of gravity. The formula is derived in the text below.

Figure 5.2b illustrates an alternative method for determination of the model point mass  $m$ . See also fig. 5.3. As one can see from the description, it is easily carried out just after the counterbalances have been adjusted. The derivation of the formula is, however, a little sophisticated. It goes as follows.

Since the merry-go-round has its centre of mass on the axis of rotation, gravity has no turning effect except for its pull  $\Delta m \cdot g$  on the extra mass  $\Delta m$ . Since this is now hung up as a mathematical pendulum in a string of length  $R$ , the model point feels a harmonic force with the force constant



**Fig. 5.3:** The merry-go-round mounted with a horizontal axis. The extra mass  $\Delta m$  has been fixed to the model point *after* the adjustment of the counterbalances. Hence it can swing back and forth as a pendulum. The period  $T$  is measured by use of the photo gate.

$$k = \frac{\Delta m \cdot g}{R} \quad (5.2)$$

The corresponding force  $-ks^2$  drives not only  $\Delta m$  back and forth, it drives  $m + \Delta m$ . The period must therefore be

$$T = 2\pi \sqrt{\frac{m + \Delta m}{k}} = 2\pi \sqrt{\frac{m + \Delta m}{\Delta m} \cdot \frac{R}{g}} \quad (5.3)$$

Solving for  $m$ , we get the result given in the text to fig. 5.2b.

Instead of determining the mass  $m$  by the theoretical formula (5.1), one can of course also do it experimentally by one of the methods from fig. 5.2. A real good exercise would be to do it by all three methods and compare the results.

#### *The balance of the merry-go-round.*

For the merry-go-round to work properly, it should be carefully balanced so that its centre of mass coincides with the axis of rotation. Had the latter been ideally vertical, and had the merry-go-round been ideally rigid, this balance would not be necessary. However, the merry-go-round is *not* ideal and when spinning quickly, an unbalance gives rise to strong forces in the bearing that might distort the structure that holds it. Furthermore, an off-axis centre of mass moves in a little circle, and if this is not strictly horizontal, the model point will seem to be moving uphill, downhill, uphill,...

To obtain a good balance, the merry-go-round is moved to a brace made for the purpose, where it is fixed firmly with a *horizontal* axis of rotation. With no model fitted, the two movable counterbalances are adjusted until the protruding arm can be left in an arbitrary direction without beginning to rotate. Actually this ideal cannot be reached completely, and a small unbalance must be accepted. This should not affect the subsequent measurements in any serious - remember that balance is not a *condition* for the theory in the ideal case. The balancing process must be repeated before the second run - only in case of a very light model can one consider to leave it out. In case of a more heavy model, balance cannot be achieved just by displacing the movable counterbalances. One must first place an extra counterbalance  $m_{\text{counter}}$  in the box opposite the protruding arm. Note that the reducing factor in (3.1) has a rather strong effect. Thus  $m_{\text{counter}} = 1\text{kg}$  only contributes to  $m$  with

$$m_{load} = \left(\frac{331mm}{958mm}\right)^2 \cdot 1000g = 119g \quad (5.3)$$

Intuitively it is fairly obvious that the equivalent mass of the model point can be determined as illustrated in the first of the two figures 5.2; i.e. in fig. 5.2a. A rigorous treatment should be founded on the theory for a rigid body's moment of inertia about an axis of rotation. This is done in appendix B.

*Some notes on the use of the extra counterbalance  $m_{counter}$*

We mention, that it is possible to use the extra counterbalance  $m_{counter}$  also in the run with the empty merry-go-round: One just has to place the movable counterbalances appropriately on the semicircle *opposite*<sup>2</sup> the box for  $m_{counter}$ . If doing so, the student can forget about  $m_{load}$  in eqn. (4.3), measure  $m = m_e$  and  $m = m_e + m_{model}$  as described in fig. 5.2 and check that the difference agrees with  $m_{model}$  determined on a pair of scales.

In case of a light model, it is possible to obtain two different determinations of  $C_d$ : One where  $m_{counter}$  is omitted in both runs and one where it is included.

In case of a light model with large air resistance, the extra  $m_{counter}$  is not necessary, but it gives a larger value of  $m$  and thereby more revolutions of the merry-go-round before it is slowed down.

---

<sup>2</sup> This would mean a rather long displacement of the movable counterbalances, and the following question arises: Does the overall geometry of merry-go-rounds geometry thereby change so much that it has a measurable effect on the air resistance?

I have not yet had the time to check this problem experimentally. One way to avoid it - and probably improve the whole design of the merry-go-round - would be to enclose the bicycle wheel, including the counterbalances, in a cylinder that rotates with it.